Thermal Energy: Different Forms of Home Heating

Physics and Mathematics of Sustainable Energy

College of the Atlantic. October 13, 2026

1. Suppose you want 100 kWh of heat to keep your house warm on a cold Maine day. If you generate this heat with a resistive electric heater:
   (a) How much electricity would you need?
   (b) How much CO_2 is released as a result?
   (c) How much would this cost in Maine?

2. If you generate this heat with a furnace that burns heating oil and the efficiency of the furnace is 80%:
   (a) How many gallons of heating oil would you need?
   (b) How much CO_2 would be released as a result?
   (c) How much would this cost in Maine?

3. If you generate this heat with a propane furnace and the efficiency of the furnace is 80%:
   (a) How many gallons of propane would you need?
   (b) How much CO_2 would be released as a result?
   (c) How much would this cost in Maine?

4. If you generate this heat with a heat pump with a COP of 3.5:
   (a) How much electricity would be required to generate this heat?
   (b) How much CO_2 would be released as a result?
   (c) How much would this cost in Maine?

- 1 kWh = 3.6 MJ = 3412 BTU
- 1 MMBTU = 1,000,000 BTU
- Calorific value of heating oil: 12.8 kWh/kg, 37.3 MJ/L, 139,000 BTU/gallon
- Calorific value of propane: 91,600 BTU/gal.
- Carbon intensity of propane: 5.75kg/gal or 215g/kWh.
- Carbon intensity of heating oil: 260g/kWh. 10.2 kg CO_2 per gallon of fuel.
- Cost of electricity in Maine \$0.32/kWh.